Module -7: Network fundamental

1- **Which of the following messages in the DHCP process are broadcasted?**

A. Request
B. Offer
C. Discover
D. Acknowledge

Answer:

- **B. Offer**
- **C. Discover**

2- **Which command would you use to ensure that an ACL does not block web-based TCP traffic?**

A. permit any

B. permit tcp any any eq 80

C. permit tcp any eq 80

D. permit any any eq tcp

Answer:
B. permit tcp any any eq 80

3) Explain Network Topologies

Answer:

- **Network topology means the physical or logical arrangement of devices in a network**
- **Common topologies are Star, Bus, Ring, Mesh, and Tree.**
- **Star topology is most commonly used in LAN networks.**
- **Topology affects network performance, cost, and reliability.**

4) Explain TCP/IP Networking Model

Answer:
TCP/IP (Transmission Control Protocol/Internet Protocol) is the standard networking model used for communication over the internet and computer networks. It defines how data is transmitted between devices.
The model has 4 layers:

1. **Application Layer – Provides services like HTTP, FTP, DNS, and Email.**
2. **Transport Layer – Uses TCP and UDP for data delivery and error checking.**
3. **Internet Layer – Handles IP addressing and routing of packets.**
4. **Network Access Layer – Manages physical transmission using cables and network hardware.**
   TCP provides reliable communication, while IP is responsible for addressing and routing data packets.

5) Explain LAN and WAN Network

Answer:

- **LAN (Local Area Network) is a network that connects devices within a small geographical area such as a home, office, school, or college. It provides high-speed communication and is usually privately owned.**
- **WAN (Wide Area Network) connects multiple LANs over large distances using public or private communication links.**
- **LAN is faster, more secure, and less costly compared to WAN. WAN is slower due to long-distance communication but allows global connectivity.**
- **Example: Office network is LAN, while the Internet is the largest WAN.**

6) Explain Operation of Switch

Answer:

- **A switch is a networking device used to connect multiple devices in a Local Area Network (LAN). It works mainly at the Data Link Layer (Layer 2) of the OSI model.**
- **The switch receives data frames and checks the destination MAC address. It then forwards the frame only to the correct device instead of broadcasting to all devices.**
- **A switch maintains a MAC Address Table to remember which device is connected to which port.**
- **This improves network performance, reduces traffic, and increases communication efficiency.**
- **Managed switches also support VLANs, security, and traffic monitoring features.**

7) Describe the purpose and functions of various network devices

Answer:
Network devices are hardware components used to establish communication and manage network traffic between computers and other devices.

- **Router: Connects different networks and forwards data packets between them using IP addresses.**
- **Switch: Connects devices within a LAN and sends data only to the intended device using MAC addresses.**
- **Hub: Connects multiple devices but broadcasts data to all connected devices.**
- **Modem: Converts digital signals to analog signals for internet communication.**
- **Access Point: Provides wireless connectivity to devices in a Wi-Fi network.**

- **Firewall: Protects the network by filtering unauthorized traffic and improving security.**

These devices help in communication, connectivity, security, and efficient data transfer.

8) Make list of the appropriate media, cables, ports, and connectors to connect switches to other devices

Answer:

1. **UTP (Unshielded Twisted Pair) Cable**
    - **Used for connecting PCs, routers, and switches.**
    - **Uses Ethernet ports and RJ45 connectors.**
    - **Commonly used in LAN networks.**
2. **STP (Shielded Twisted Pair) Cable**
    - **Similar to UTP but provides better protection from interference.**
    - **Uses RJ45 connectors.**
    - **Used in high-interference environments.**
3. **Fiber Optic Cable**
    - **Used for high-speed and long-distance communication.**
    - **Uses SFP ports with LC or SC connectors.**
    - **Provides faster data transmission than copper cables.**
4. **Coaxial Cable**
    - **Used in cable internet and older network systems.**
    - **Uses BNC connectors and coaxial ports.**
    - **Resistant to signal loss and interference.**
5. **Console Cable**
    - **Used to configure and manage switches or routers.**
    - **Connects through the Console port using RJ45 or USB connectors.**
    - **Mainly used by network administrators.**

6. **Wireless Media**
    - **Uses radio waves instead of physical cables.**
    - **Used with wireless access points and Wi-Fi devices.**
    - **Helps devices connect without wired connections.**

9) Define Network devices and hosts

Answer:

- **Network devices are hardware components that help in transmitting, managing, and controlling network communication. Examples include routers, switches, hubs, firewalls, and modems.**
- **These devices help in routing data, connecting systems, providing security, and managing traffic flow in the network.**
- **Hosts are end-user devices connected to the network that can send or receive data. Examples include computers, laptops, mobile phones, printers, and servers.**
  Each host is assigned an IP address for identification and communication in the network.
- **Hosts use network devices to communicate with other devices locally or over the internet.**